

Xueyi Zhang

Research Fellow
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Profile

Affiliation

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Research Achievements

17 **Publications in**

ICML, NeurIPS, ICCV, ACM Multimedia, ACL, SIGIR, and AAAI
Including **5 first-author oral papers**.

25 **Published patent applications**

20 granted
5 in substantive examination

19 **Competition honors**

Academic Service

Area Chair
AAAI 2027 • Canada
ACM MM 2026 • Brazil

Workshop Organizer
The Web Conference (WWW) 2027 • Dublin, Ireland

Workshop Chair
The Innovation Conference on Informatics 2026 • Shenzhen, China

Special Session Organizer
ICASSP 2027 • Canada

Organizing Committee
Nanyang Speech Forum 2025 & 2026 • Singapore

Research Experience

Research Fellow – National University of Singapore

Jun 2026 – Present

School of Computing, Singapore

- Working with Jiaheng Zhang and Bingsheng He (IEEE Fellow).
- Collaborating with Haizhou Li (IEEE Fellow; Singapore Academy of Engineering Fellow) and Nitin Chawla (ST Fellow; Director of Edge AI, ST).

Research Assistant – The Chinese University of Hong Kong, Shenzhen

Mar 2023 – Jun 2026

Working with Prof. Haizhou Li.

Recent News

24 Aug 2026 – HELIX Corporate Lab Launch

Represented the HELIX Lab team in presenting research findings to the **NUS President Prof. Tan Eng Chye**, **Singapore's Senior Minister of State for Trade and Industry Ms Low Yen Ling**, and **STMicroelectronics Executive Vice President Bertrand Stoltz**. Featured on NUS's LinkedIn and Instagram.

Demo: Real-Time AI-Powered Translation Glasses. Deployed large AI models directly on the glasses for **real-time, any-to-any translation across 14 languages**, with seamless language switching and **latency of at most 200 ms—no internet connection required**.

Demo Co-leads: Prof. Haizhou Li, Prof. Massimo Alioto, and **Dr. Xueyi Zhang**.



Lab launch



Presentation
Xueyi Zhang (far right)



14-language translation demo

Project Experience

Full-Duplex Affective Dialogue Robot

Dec 2024 – Jun 2026 CUHK-Shenzhen

Working with Prof. Haizhou Li.

- Inspired by Daniel Kahneman's *Thinking, Fast and Slow*, designed fast/slow responses for **human-like, full-duplex dialogue**: the robot listens while speaking and supports natural interruptions.
- Combines speech, facial expressions, gaze, and posture to understand emotion and coordinate empathetic speech, expressive faces, and head/gaze tracking.

Related manuscript: *Full-Duplex Affective Dialogue Robot* (**under review**).

Education

National University of Defense Technology

Ph.D. in Control Science and Engineering 2020–2025

- **CAST Young Talent Support Program (Doctoral):** Inaugural cohort.
- **Deya Scholar:** Highest student honor; only 10/year university-wide.
- **National Scholarship for Doctoral Students:** Awarded twice.
- **Outstanding Student Leader.**

Dalian Maritime University

B.Eng. in Electronic Information Engineering 2016–2020

- **President's Scholarship:** Highest university student honor.
- **Top Ten Students Award:** Only 10 recipients/year university-wide.
- **Outstanding Student Leader.**

Selected Publications

First-author works supporting the proposed wearable-AI project (WP1–WP4).

1. **Generalization-Preserved Learning: Closing the Backdoor to Catastrophic Forgetting in Continual Deepfake Detection.**
   Xueyi Zhang et al. *ICCV 2025*. [Paper]
   Experience preserving generalization under changing data informs event representation and evidence retention (WP1/WP3).
2. **Choose Your Expert: Uncertainty-Guided Expert Selection for Continual Deepfake Detection.**
   Xueyi Zhang et al. *ACM Multimedia 2025*. **Oral**. [Paper]
   Uncertainty-guided expert selection motivates query-conditioned refinement and runtime compute allocation (WP2/WP4).
3. **TrustCLIP: Learning from Noisy Labels via Semantic Label Verification and Trust-aligned Gradient Projection.**
   Xueyi Zhang et al. *ACM Multimedia 2025*. **Oral**. [Paper]
   Learning with noisy labels informs robust teacher-supervised training of event scorers and refinement policies (WP1/WP2).
4. **Language Without Borders: A Dataset and Benchmark for Code-Switching Lip Reading.**
   Xueyi Zhang et al. *NeurIPS 2024*. [Paper]
   Multilingual visual-speech and benchmarking experience supports multimodal workload and evaluation design (WP4).
5. **Listening to the Brain: Multi-Band sEEG Auditory Reconstruction via Dynamic Spatio-Temporal Hypergraphs.**
   Xueyi Zhang et al. *NeurIPS 2025*. [Paper]
   Experience modeling complex signals and spatio-temporal relations informs cross-modal event representation (WP1).
6. **Rep Deep & Machine Learning: Exemplar-Free Continual Video Action Recognition via Slow-Fast Collaborative Learning.**
   Xueyi Zhang et al. *AAAI 2026*. **Oral**. [Paper]
   Deep/analytic learning and dynamic branch fusion motivate budget-aware selective refinement (WP2/WP4).
7. **Slow-Fast Time Parameter Aggregation Network for Class-Incremental Lip Reading.**
   Xueyi Zhang et al. *ACM Multimedia 2023*, pp. 747–756. **Oral**. [Paper]
   Continual lip-reading experience informs adaptive visual-speech representations for wearable AI (WP1/WP2).
8. **EventLip: Enhancing Event-Based Lip Reading via Frequency-Aware Spatiotemporal Hypergraph Modeling.**
   Xueyi Zhang et al. *ACM Multimedia 2025*. [Paper]
   Event-based visual-speech modeling informs event representations for wearable perception (WP1).

Competition Awards

19 awards and distinctions: 12 national, 3 provincial, 2 in international competitions, 1 regional, and 1 municipal.

No.	Competition / Track	Scope	Award	Role / Position
1	iCAN Innovation & Entrepreneurship	National	**Top Five Finalist**	Captain
2	iCAN Innovation & Entrepreneurship	National	**First Prize**	Captain
3	China Internet+ Innovation & Entrepreneurship	National	**Bronze**	Captain
4	Underwater Robotics: Fixed-Point Traversal	International	**Third Prize**	Captain
5	Underwater Robotics: Fixed-Point Grasping	International	**Third Prize**	Captain
6	China Student Computer Design: AI	National	**Second Prize**	Team pos. 2
7	China Student Computer Design: AI	National	**Second Prize**	Team pos. 3
8	China Student Computer Design: Visualization	National	**Third Prize**	Team pos. 3
9–10	Huawei Cup Intelligent Design	National	**Third Prize (2)**	Captain
11–12	Energy Conservation & Emission Reduction	National	**Third Prize (2)**	Captain
13	Challenge Cup: Academic Science & Technology	Provincial	**Grand Prize**	Captain
14	China Internet+ Innovation & Entrepreneurship	Provincial	**Gold**	Captain
15	iCAN Innovation & Entrepreneurship	Provincial	**First Prize**	Captain
16	CCCC Network Technology (Central China)	Regional	**Third Prize**	Captain
17	Changsha Innovation & Entrepreneurship (PG)	Municipal	**First Prize**	Captain
18	Huawei Cup Graduate Mathematical Modeling	National	**Second Prize**	Captain
19	Teddy Cup Data Analysis Skills	National	**Second Prize**	Captain